Probability

Theoretical population models

How can we study a random process?

Modeling: Make assumptions about its structure (classical statistics)

Analysis: with pure math and probability, if the process is known

Simulation: sample from the process a million bajillion times, see what patterns emerge

Population models are called **random variables (RVs)**.



RVs are an object in probability theory in general.

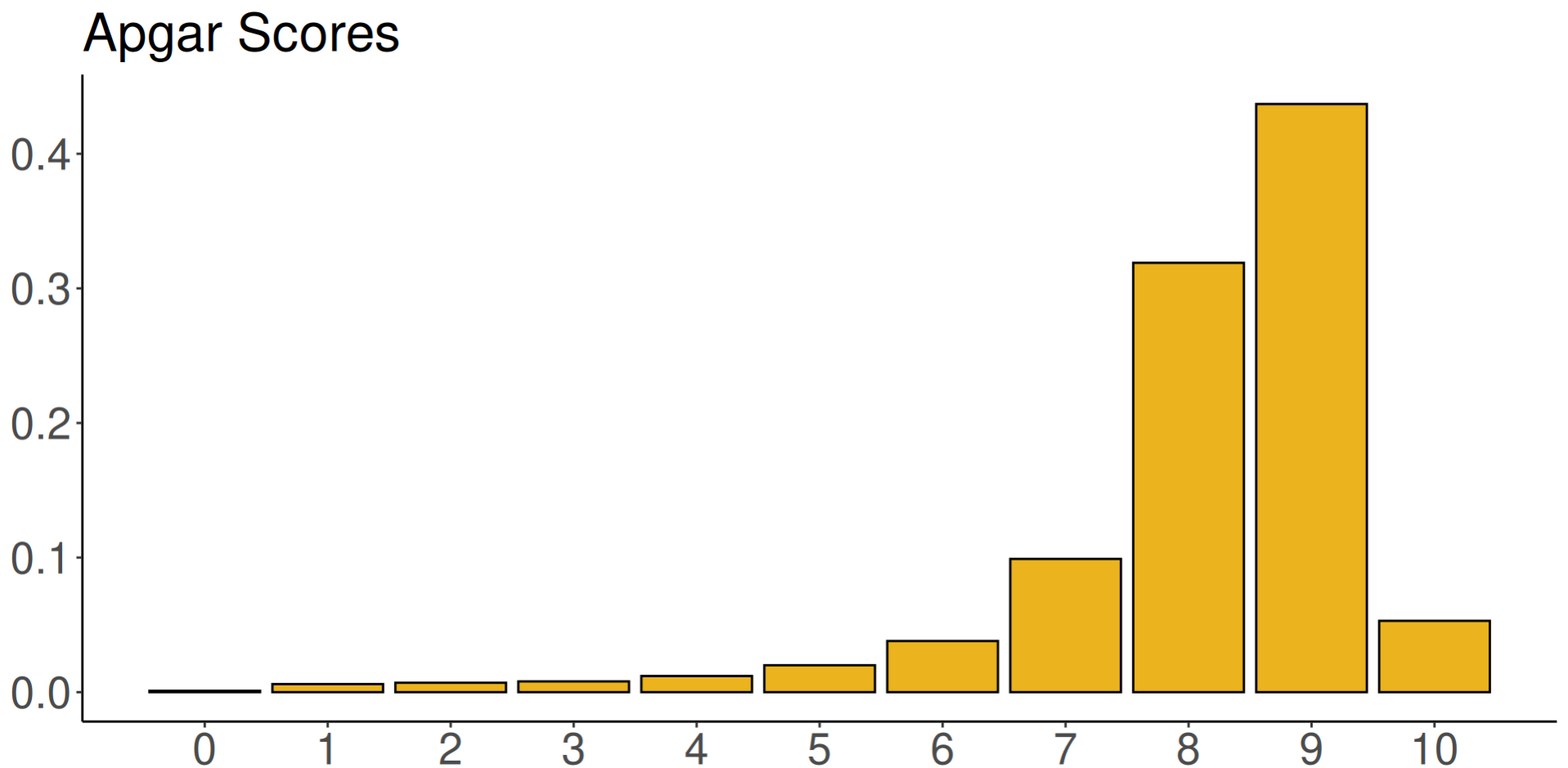
More specifically, an RV *quantifies* a random process.

- Outcome of 3 coin flips: not an RV
- Number of heads in 3 coin flips: is an RV

A random variable generates numeric data, e.g. penguin flipper lengths:

181, 196, 195, 193, 190, 181, …

The Apgar score for newborns is on a scale from 0-10 based on condition immediately after birth.



An RV X has a **support** of possible values.

- X = process, x = specific outcome

x	0	1	2	3	4	5	6	7	8	9	10

But how likely is each value?

A **probability distribution** assigns a probability to each value.

x	**0**	**1**	**2**	**3**	**4**	**5**
$P(X = x)$	0.001	0.006	0.007	0.008	0.012	0.02

x	**6**	**7**	**8**	**9**	**10**
$P(X = x)$	0.038	0.099	0.319	0.437	0.053

This is a population, i.e. a process for generating data.

Exercises | Solutions/Hints

Let random variable X be the total number of heads in 3 coin flips. X has support {0, 1, 2, 3}.

- What is the probability distribution of X?
- Use R to simulate 10000 instances/draws from X, and graph the results. Does it match what you expect?



There are two broad categories of random variable.

A **discrete** RV can only take on specific values. Apgar score is discrete.

- The probabilities sum to 1
- Values “in between” the support are impossible

The probability function of a discrete RV is called a **probability mass function** (pmf).

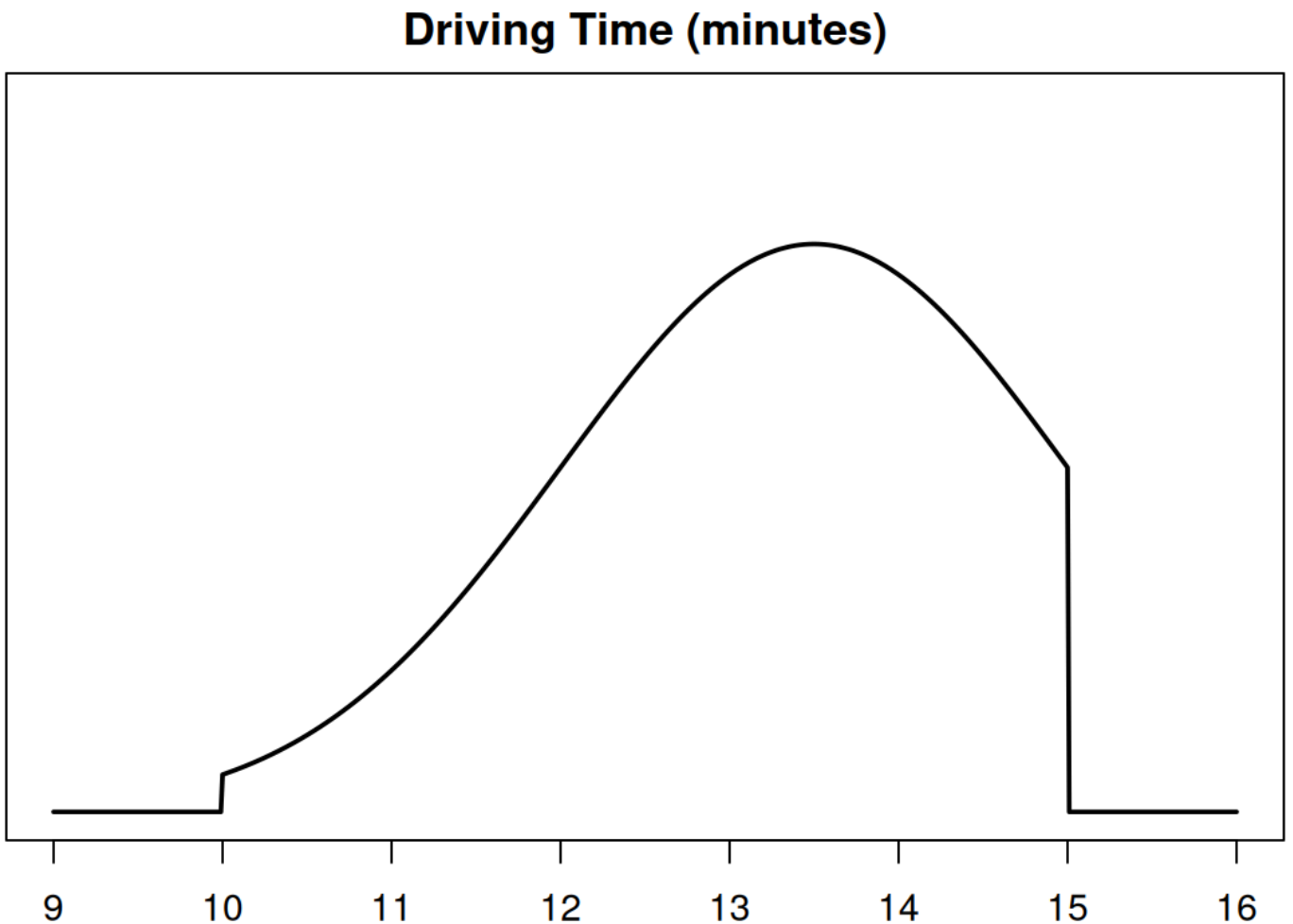
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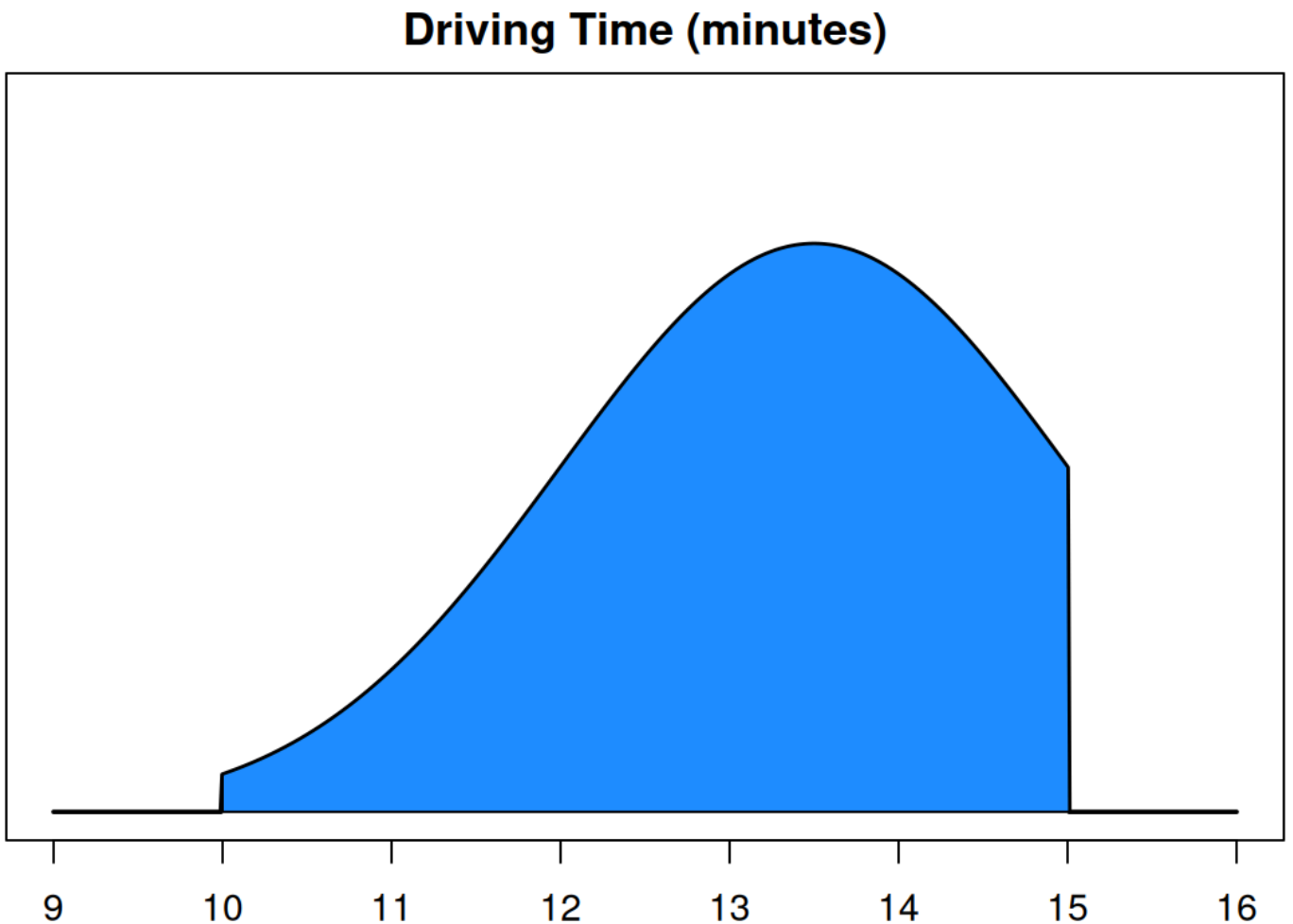
A **continuous** RV can take on any value in a range, e.g. time, distance, height.

A continuous RV is drawn as a smooth curve, called a **probability density function** (pdf).

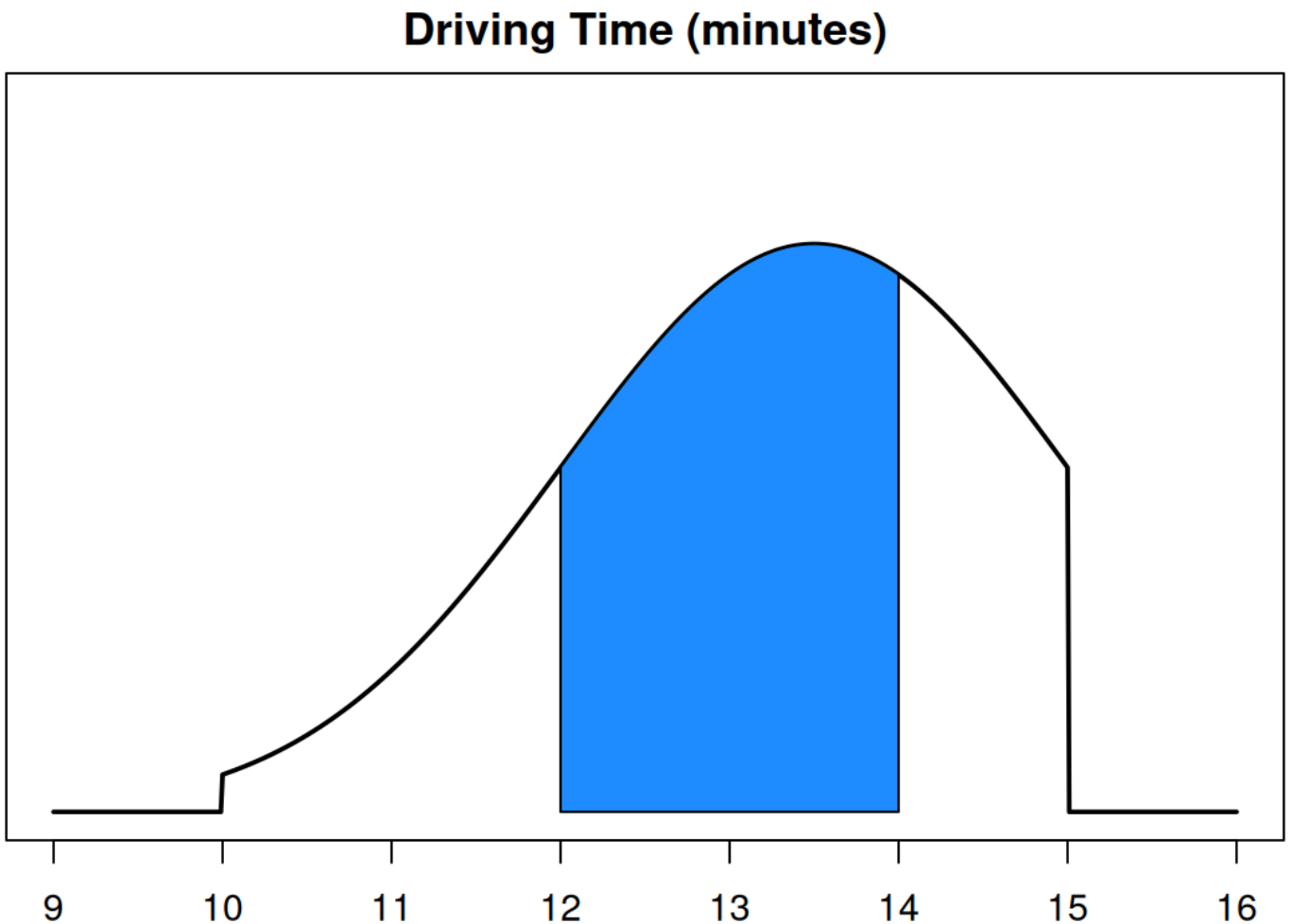
It takes 10 to 15 minutes to drive to class.



Probabilities are the area *under* the pdf. The total area is 1.



$P(12 \leq \text{driving time } \leq 14)$:



The probability of a single value is 0, e.g.
$P(\text{driving time } = 12)$.

Exercises | Solutions/Hints

Are the following variables discrete or continuous?

- Number of tornadoes in Wisconsin in a year
- Inches of rainfall
- Age at college graduation





RVs *quantify* a random process/population, allowing us to calculate summaries

- Expected value: population mean
- Variance: population variance

What is the mean or *typical* value in a population?

Each person in a population has a value of 0 or 1.
Is the average (0+1)/2 = 0.5?

What if I told you that this is the pmf?

x	**0**	**1**
$P(X = x)$	0.2	0.8

The mean of a RV is weighted according to the probabilities.

$$E(X) \;=\; 0(0.2) + 1(0.8) \;=\; 0.8$$

In general the mean of a discrete RV is

$$E(X) \;=\; \mu \;=\; \sum_{x} x \cdot P(X = x)$$

This is the **expected value** or **expectation** (population mean).

We can also quantify the spread of a population. The variance of a RV is the average squared distance from the values to the mean.

Start with $[x - E(X)]^2$.

Again, take a weighted average.

The variance of a **discrete** RV is

$$V(X) \;=\; \sigma^2 \;=\; \sum_x [x - E(X)]^2 \cdot P(X = x)$$

The **standard deviation** is the square root of the variance.

$$sd(X) \;=\; \sigma \;=\; \sqrt{V(X)}$$

$E(X)$ and $V(X)$ are calculated with integration for continuous RVs, but have the same interpretation as the discrete case:

- $E(X)$: population mean
- $V(X)$: population variance

Exercises | Solutions/Hints

Consider an unfair die:

x	1	2	3	4	5	6
$P(X = x)$	0.1	0.1	0.1	0.2	0.2	0.3

- Calculate $E(X)$ and $V(X)$ with R.
- Simulate 100 "games" where player A rolls a fair die and player B rolls the unfair die. Count the wins and ties.
- If you are player A, how might you try to prove that player B is cheating?



Looking ahead

In chapter 9, we'll learn about specific types of distributions, the *binomial* and *normal*. If you want a head start, re-consider the simulation of the number of heads out of 3 coin flips.

```
heads <- replicate(10000,
           sum(sample(0:1, size = 3, replace = T)))
```

What happens to this distribution as the total number of coin flips increases? What happens when you switch to an unfair coin? Experiment with a few different values and plot histograms of the results.